

Міністерство освіти і науки України
Національний Університет “Львівська Політехніка”



Лабораторна робота
з дисципліни
“Об'єктно-орієнтоване програмування, частина 2”

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Лабораторна робота 3.

Тема: Робота з табличними даними.

Мета: Засвоїти основи роботи з бібліотекою Pandas, навчитися працювати з табличними даними, виконувати фільтрацію, сортування, групування та застосовувати функції для аналітики.

Завдання :

```
import pandas as pd
```

Завдання 1. Завантаження файлу

```
try:
```

```
    from google.colab import files
```

```
    print("📁 Завантажте Job_opportunities.csv:")
```

```
    uploaded = files.upload()
```

```
    filepath = list(uploaded.keys())[0]
```

```
except ImportError:
```

```
    filepath = "Job_opportunities.csv"
```

```
df = pd.read_csv(filepath)
```

```
print("✅ Дані завантажено!\n")
```

Завдання 2. Первинний аналіз

```
print("=== ПЕРШІ 5 РЯДКІВ ===")
```

```
print(df.head(), "\n")
```

```
print("=== ОСТАННІ 5 РЯДКІВ ===")
```

```
print(df.tail(), "\n")
```

```
rows, cols = df.shape
```

```
print(f"▲ Розмір датасету: {rows} рядків × {cols} стовпців")
```

```
print(f"📄 Пам'ять: {df.memory_usage(deep=True).sum()/1024:.2f} КБ\n")
```

```
print("=== ТИПИ ДАНИХ ===")
```

```
print(df.dtypes, "\n")
```

```
print("=== ІНФО ===")
```

```
df.info()
```

```
missing = df.isnull().sum()
print("\n=== ПРОПУЩЕНІ ЗНАЧЕННЯ ===")
print(missing)
if missing.sum() == 0:
    print("✅ Пропущених значень немає\n")
else:
    print(f"⚠️ Пропущено {missing.sum()} значень\n")
```

Завдання 3. Фільтрація вакансій

```
print("\n=== ВАКАНСІЇ З ГАЛУЗІ 'Cloud Computing' ===")
cloud_jobs = df[df["Industry"] == "Cloud Computing"]
print(cloud_jobs[["Job Title", "Industry", "Location", "Salary Range"]].to_string(index=False))

print("\n=== ВАКАНСІЇ РІВНЯ 'Senior' ===")
senior_jobs = df[df["Experience Level"] == "Senior"]
print(senior_jobs[["Job Title", "Experience Level", "Salary Range"]].to_string(index=False))

print("\n=== FULL-TIME В ЛОНДОНІ ===")
ft_london = df[(df["Job Type"] == "Full-Time") & (df["Location"] == "London")]
print(ft_london[["Job Title", "Job Type", "Location", "Salary Range"]].to_string(index=False))
```

Завдання 4. Сортування за зарплатою

```
df["Max Salary"] = df["Salary Range"].str.extract(r"£([\d,]+)\s*$")[0].str.replace(",", "").astype(int)

print("\n=== ТОП-5 НАЙВИЩИХ ЗАРПЛАТ ===")
top5 = df.sort_values(by="Max Salary", ascending=False).head(5)
print(top5[["Job Title", "Salary Range", "Max Salary"]].to_string(index=False))
```

Завдання 5. Групування та середня мінімальна зарплата

```
df["Min Salary"] = df["Salary Range"].str.extract(r"£([\d,]+)")[0].str.replace(",", "", "").astype(int)
grouped = df.groupby("Industry").agg(
    vacancies=("Job Title", "count"),
    avg_min_salary=("Min Salary", "mean")
).sort_values(by="avg_min_salary", ascending=False)
print("\n=== СЕРЕДНЯ МІНІМАЛЬНА ЗАРПЛАТА ЗА ГАЛУЗЯМИ ===")
print(grouped)

top_industry = grouped["avg_min_salary"].idxmax()
print(f"\n🏆 Галузь з найвищою середньою мінімальною зарплатою: {top_industry} (£{grouped.loc[top_industry, 'avg_min_salary']:.0f})")
```

Завдання 6. Категорії зарплати

```
def salary_cat(s):
    if s <= 40000:
        return "Low"
    elif s <= 70000:
        return "Medium"
    else:
        return "High"

df["Salary Category"] = df["Max Salary"].apply(salary_cat)
print("\n=== РОЗПОДІЛ ЗА КАТЕГОРІЯМИ ЗАРПЛАТ ===")
print(df["Salary Category"].value_counts())
```

Завдання 7. Часовий аналіз ринку вакансій

```
df["Date Posted"] = pd.to_datetime(df["Date Posted"], format="%m/%d/%Y")
df["Year"] = df["Date Posted"].dt.year
by_year = df.groupby("Year").agg(vacancies=("Job Title", "count")).sort_index()
print("\n=== КІЛЬКІСТЬ ВАКАНСІЙ ЗА РОКАМИ ===")
print(by_year)
```

```
best_year = by_year["vacancies"].idxmax()
print(f'\n🐼 Найактивніший рік: {best_year} ({by_year.loc[best_year,
'vacancies']}) вакансій")
```

Результат:

```
*** 📁 Завантажте Job_opportunities.csv:
Вибрати файли Job_opportunities.csv
Job_opportunities.csv(text/csv) - 56360 bytes, last modified: 01.03.2026 - 100% done
Saving Job_opportunities.csv to Job_opportunities (2).csv
✅ Дані завантажено!
```

=== ПЕРШІ 5 РЯДКІВ ===

	Job Title	Job Description	Required Skills	\
0	Software Engineer	Develop software	Java, Python	
1	Data Analyst	Analyze data	SQL, Excel	
2	Network Engineer	Maintain networks	Cisco, WAN	
3	Cloud Architect	Design cloud	AWS, Azure	
4	Cybersecurity Analyst	Protect data	Cybersecurity	

	Salary Range	Location	Company	Experience Level	\
0	£40,000 - £60,000	London	ABC Tech	Entry-Level	
1	£35,000 - £50,000	Manchester	XYZ Analytics	Junior	
2	£45,000 - £70,000	Birmingham	Network Solutions	Mid-Level	
3	£60,000 - £90,000	Edinburgh	Cloud Innovators	Senior	
4	£50,000 - £80,000	Glasgow	SecureGuard	Mid-Level	

	Industry	Job Type	Date Posted
0	Technology	Full-Time	1/5/2023
1	Analytics	Full-Time	2/10/2023
2	Networking	Full-Time	3/15/2023
3	Cloud Computing	Full-Time	4/20/2023
4	Security	Full-Time	5/25/2023

=== ОСТАННІ 5 РЯДКІВ ===

	Job Title	Job Description	\
359	AI Ethics Consultant	Ensure AI ethics	
360	IT Support Specialist	Provide IT support	
361	Data Analytics Manager	Manage data analytics	
362	IT Project Coordinator	Coordinate IT projects	
363	Cloud Solutions Analyst	Analyze cloud solutions	

	Required Skills	Salary Range	Location	\
359	Ethics, AI Governance	£75,000 - £120,000	Leeds	

```

... 360      Windows, Troubleshooting  £45,000 - £75,000  Sheffield
361 Data Analytics, Team Management £80,000 - £130,000 Nottingham
362      Project Coordination  £50,000 - £85,000  Liverpool
363      AWS, Azure, Cloud Analysis  £70,000 - £110,000 Birmingham

```

```

      Company Experience Level      Industry  Job Type \
359      AI Ethics      Senior      Ethics  Full-Time
360 IT Support Specialists  Mid-Level      Support  Full-Time
361      Data Managers      Senior      Analytics  Full-Time
362 Project Coordinators  Mid-Level  Project Management  Full-Time
363      Cloud Analysts  Mid-Level      Cloud Computing  Full-Time

```

```

      Date Posted
359  5/26/2019
360  6/21/2019
361  7/16/2019
362  8/11/2019
363  9/6/2019

```

 Розмір датасету: 364 рядків x 10 стовпців
 Пам'ять: 229.26 KB

```

=== ТИПИ ДАНИХ ===

```

```

Job Title      object
Job Description object
Required Skills object
Salary Range   object
Location       object
Company        object
Experience Level object
Industry       object
Job Type       object
Date Posted    object
dtype: object

```

```

=== ІНФО ===

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 364 entries, 0 to 363

```

```
... RangeIndex: 364 entries, 0 to 363
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   Job Title              364 non-null    object
1   Job Description        364 non-null    object
2   Required Skills        364 non-null    object
3   Salary Range          364 non-null    object
4   Location               364 non-null    object
5   Company               364 non-null    object
6   Experience Level       364 non-null    object
7   Industry               364 non-null    object
8   Job Type               364 non-null    object
9   Date Posted           364 non-null    object
dtypes: object(10)
memory usage: 28.6+ KB
```

```
=== ПРОПУЩЕНІ ЗНАЧЕННЯ ===
Job Title          0
Job Description    0
Required Skills    0
Salary Range      0
Location          0
Company           0
Experience Level   0
Industry          0
Job Type          0
Date Posted       0
dtype: int64
✅ Пропущених значень немає
```

```
=== ВАКАНСІЇ З ГАЛУЗІ 'Cloud Computing' ===
      Job Title      Industry  Location  Salary Range
Cloud Architect  Cloud Computing  Edinburgh  £60,000 - £90,000
Cloud Solutions Architect  Cloud Computing  Newcastle  £65,000 - £100,000
Cloud Solutions Analyst  Cloud Computing  Leeds  £45,000 - £75,000
Cloud Migration Specialist  Cloud Computing  Glasgow  £55,000 - £90,000
```

```

... Cloud Solutions Architect Cloud Computing Manchester £60,000 - £90,000
Cloud Solutions Architect Cloud Computing Cardiff £60,000 - £90,000
Cloud Solutions Analyst Cloud Computing Manchester £45,000 - £75,000
Cloud Architect Cloud Computing Edinburgh £60,000 - £90,000
Cloud Solutions Architect Cloud Computing London £60,000 - £90,000
Cloud Solutions Analyst Cloud Computing Manchester £45,000 - £75,000
Cloud Architect Cloud Computing Edinburgh £60,000 - £95,000
Cloud Architect Cloud Computing Edinburgh £70,000 - £110,000
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Cloud Architect Cloud Computing Edinburgh £70,000 - £110,000
Cloud Solutions Analyst Cloud Computing Manchester £55,000 - £90,000
Cloud Solutions Analyst Cloud Computing Birmingham £70,000 - £110,000
Cloud Architect Cloud Computing Edinburgh £70,000 - £110,000
Cloud Solutions Analyst Cloud Computing Manchester £55,000 - £90,000
Cloud Solutions Analyst Cloud Computing Birmingham £70,000 - £110,000

```

```

=== БАКАНСІЇ ПІБНЯ 'Senior' ===

```

	Job Title	Experience Level	Salary Range
	Cloud Architect	Senior	£60,000 - £90,000
	IT Project Manager	Senior	£70,000 - £100,000
	Cloud Solutions Architect	Senior	£65,000 - £100,000
	IT Consultant	Senior	£60,000 - £90,000
	IT Sales Manager	Senior	£70,000 - £110,000
	Software Architect	Senior	£75,000 - £120,000
	IT Procurement Manager	Senior	£70,000 - £110,000
	Data Analytics Manager	Senior	£70,000 - £110,000
	IT Governance Manager	Senior	£65,000 - £100,000
	IT Change Manager	Senior	£60,000 - £100,000
IT Business Continuity Manager		Senior	£70,000 - £110,000
	Cloud Solutions Architect	Senior	£60,000 - £90,000
	IT Project Manager	Senior	£70,000 - £100,000
	Cloud Solutions Architect	Senior	£60,000 - £90,000

...	Cloud Solutions Architect	Senior	£60,000 - £90,000
	IT Consultant	Senior	£60,000 - £90,000
	IT Sales Manager	Senior	£70,000 - £110,000
	Software Architect	Senior	£75,000 - £120,000
	IT Procurement Manager	Senior	£70,000 - £110,000
	Data Analytics Manager	Senior	£70,000 - £110,000
	IT Governance Manager	Senior	£65,000 - £100,000
	Cloud Architect	Senior	£60,000 - £90,000
	IT Project Manager	Senior	£70,000 - £100,000
	Cloud Solutions Architect	Senior	£60,000 - £90,000
	IT Consultant	Senior	£60,000 - £90,000
	IT Sales Manager	Senior	£70,000 - £110,000
	Software Architect	Senior	£75,000 - £120,000
	IT Procurement Manager	Senior	£70,000 - £110,000
	Data Analytics Manager	Senior	£70,000 - £110,000
	IT Governance Manager	Senior	£65,000 - £100,000
	Cloud Architect	Senior	£60,000 - £95,000
	IT Project Manager	Senior	£70,000 - £110,000
	Cloud Architect	Senior	£70,000 - £110,000
	IT Project Manager	Senior	£75,000 - £120,000
	IT Governance Manager	Senior	£80,000 - £130,000
	Cloud Architect	Senior	£70,000 - £110,000
	IT Project Manager	Senior	£75,000 - £120,000
	IT Governance Manager	Senior	£80,000 - £130,000
	IT Consultant	Senior	£75,000 - £120,000
	IT Sales Manager	Senior	£80,000 - £130,000
	Software Architect	Senior	£90,000 - £140,000
	IT Procurement Manager	Senior	£75,000 - £120,000
	AI Ethics Consultant	Senior	£75,000 - £120,000
	Data Analytics Manager	Senior	£80,000 - £130,000
	Cloud Architect	Senior	£70,000 - £110,000
	IT Project Manager	Senior	£75,000 - £120,000
	IT Governance Manager	Senior	£80,000 - £130,000
	IT Consultant	Senior	£75,000 - £120,000
	IT Sales Manager	Senior	£80,000 - £130,000
	Software Architect	Senior	£90,000 - £140,000
	IT Procurement Manager	Senior	£75,000 - £120,000
	AI Ethics Consultant	Senior	£75,000 - £120,000

...	AI Ethics Consultant	Senior	£75,000 - £120,000
	Data Analytics Manager	Senior	£80,000 - £130,000
	Cloud Architect	Senior	£70,000 - £110,000
	IT Project Manager	Senior	£75,000 - £120,000
	IT Governance Manager	Senior	£80,000 - £130,000
	IT Consultant	Senior	£75,000 - £120,000
	IT Sales Manager	Senior	£80,000 - £130,000
	Software Architect	Senior	£90,000 - £140,000
	IT Procurement Manager	Senior	£75,000 - £120,000
	AI Ethics Consultant	Senior	£75,000 - £120,000
	Data Analytics Manager	Senior	£80,000 - £130,000
	Cloud Architect	Senior	£70,000 - £110,000
	IT Project Manager	Senior	£75,000 - £120,000
	IT Governance Manager	Senior	£80,000 - £130,000
	IT Consultant	Senior	£75,000 - £120,000
	IT Sales Manager	Senior	£80,000 - £130,000
	Software Architect	Senior	£90,000 - £140,000
	IT Procurement Manager	Senior	£75,000 - £120,000
	AI Ethics Consultant	Senior	£75,000 - £120,000
	Data Analytics Manager	Senior	£80,000 - £130,000

=== FULL-TIME В ЛОНДОНЕ ===

Job Title	Job Type	Location	Salary Range
Software Engineer	Full-Time	London	£40,000 - £60,000
Database Analyst	Full-Time	London	£40,000 - £65,000
Business Analyst	Full-Time	London	£45,000 - £80,000
Cloud Support Engineer	Full-Time	London	£40,000 - £70,000
IT Trainer Assistant	Full-Time	London	£30,000 - £50,000
UX Designer	Full-Time	London	£45,000 - £75,000
IT Asset Manager	Full-Time	London	£55,000 - £90,000
Software Development Intern	Full-Time	London	£20,000 - £30,000
UX/UI Designer	Full-Time	London	£40,000 - £70,000
Cloud Support Engineer	Full-Time	London	£40,000 - £70,000
IT Compliance Officer	Full-Time	London	£60,000 - £95,000
IT Project Coordinator	Full-Time	London	£40,000 - £70,000
Software Engineer	Full-Time	London	£50,000 - £80,000
IT Support Analyst	Full-Time	London	£30,000 - £45,000
Cloud Solutions Architect	Full-Time	London	£60,000 - £90,000
Cloud Support Engineer	Full-Time	London	£40,000 - £70,000

```

...   Cloud Solutions Architect Full-Time   London £60,000 - £90,000
      Cloud Support Engineer Full-Time     London £40,000 - £70,000
      IT Compliance Officer Full-Time      London £60,000 - £95,000
      IT Project Coordinator Full-Time     London £40,000 - £70,000
      Software Developer Full-Time         London £45,000 - £75,000
      Full-stack Developer Full-Time       London £50,000 - £85,000
      Full-stack Developer Full-Time       London £50,000 - £85,000
      DevOps Engineer Full-Time            London £55,000 - £90,000
      Full-stack Developer Full-Time       London £50,000 - £85,000
      DevOps Engineer Full-Time            London £55,000 - £90,000
      Full-stack Developer Full-Time       London £50,000 - £85,000
      DevOps Engineer Full-Time            London £55,000 - £90,000
      Full-stack Developer Full-Time       London £50,000 - £85,000
      DevOps Engineer Full-Time            London £55,000 - £90,000

```

=== ТОП-5 НАЙВИЩИХ ЗАРПЛАТ ===

Job Title	Salary Range	Max Salary
Software Architect	£90,000 - £140,000	140000
Software Architect	£90,000 - £140,000	140000
Software Architect	£90,000 - £140,000	140000
Software Architect	£90,000 - £140,000	140000
Data Analytics Manager	£80,000 - £130,000	130000

=== СЕРЕДНЯ МІНІМАЛЬНА ЗАРПЛАТА ЗА ГАЛУЗЯМИ ===

	vacancies	avg_min_salary
Industry		
Software Design	7	83571.428571
Governance	9	71666.666667
Sales	8	71250.000000
Business Continuity	1	70000.000000
Consulting	7	68571.428571
Ethics	7	68571.428571
Analytics	9	65000.000000
Artificial Intelligence	7	63571.428571
Data Science	9	61666.666667
Mobile Development	7	61428.571429
Compliance	8	61250.000000
Project Management	16	60937.500000
Product Development	25	60000.000000

Business Continuity	1	70000.000000
... Consulting	7	68571.428571
Ethics	7	68571.428571
Analytics	9	65000.000000
Artificial Intelligence	7	63571.428571
Data Science	9	61666.666667
Mobile Development	7	61428.571429
Compliance	8	61250.000000
Project Management	16	60937.500000
Cloud Computing	25	60800.000000
Blockchain	7	60714.285714
Machine Learning	1	60000.000000
Change Management	1	60000.000000
Procurement	18	59722.222222
Data Privacy	7	57857.142857
Security	28	57321.428571
Data Engineering	8	55625.000000
Asset Management	1	55000.000000
DevOps	10	55000.000000
Audit	7	52857.142857
Business Analysis	10	52500.000000
Risk Management	7	50000.000000
Networking	9	48888.888889
Web Development	22	47272.727273
Software Development	9	46111.111111
Database	7	45714.285714
IT Services	3	45000.000000
Design	17	44823.529412
Quality Assurance	15	44000.000000
Data Analysis	7	43571.428571
Technology	1	40000.000000
IT Analysis	1	40000.000000
Support	30	38266.666667
Education	15	35666.666667
Research	1	30000.000000
Training	1	25000.000000
Cloud	1	20000.000000

👉 Галузь з найвищою середньою мінімальною зарплатою: Software Design (£83571)

=== РОЗПОДІЛ ЗА КАТЕГОРІЯМИ ЗАРПЛАТ ===

Salary Category

High 296

Medium 60

Low 8

Name: count, dtype: int64

=== КІЛЬКІСТЬ ВАКАНСІЙ ЗА РОКАМИ ===

vacancies

Year

2019 44

2020 44

2021 106

2022 49

2023 121

🔥 Найактивніший рік: 2023 (121 вакансій)

Висновок:

У процесі виконання лабораторної роботи було здійснено комплексний первинний аналіз датасету вакансій, проведено фільтрацію записів за галуззю, рівнем кваліфікації та географічним розташуванням. У результаті групування даних вдалося визначити галузь із найвищим середнім рівнем мінімальної заробітної плати, а також встановити розподіл вакансій відповідно до зарплатних категорій. Часовий аналіз підтвердив, що інтенсивність публікацій вакансій на ринку праці суттєво варіювалась залежно від року, досягаючи максимального значення у найбільш активний період для ІТ-спеціальностей.